WORKED SOLUTION

STEP I 2019, Question 2

Pure • Tangents to Parametric Curves

STEP I 2019, Question 2

Read the question before you go on. It is in the STEP I paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

The parametrisation is chosen so that the gradient at parameter t is simply t, which makes every tangent line very clean. Finding where two tangents meet is then a matter of factorising a difference of cubes, and the perpendicularity condition $pq = -1$ lets us rewrite everything in terms of the sum $p + q$. For the common points, substituting one parametrisation into the other gives a polynomial with a repeated root, and that repeated root is telling us something about the geometry.

Solution

The tangent

With $x = 3t^2$ and $y = 2t^3$,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y/\mathrm{d}t}{\mathrm{d}x/\mathrm{d}t} = \frac{6t^2}{6t} = t \qquad (t \neq 0).$$

So the tangent at $(3p^2, 2p^3)$ has gradient p and equation

$$y - 2p^3 = p\,(x - 3p^2) \quad \Longrightarrow \quad y = px - 3p^3 + 2p^3 = px - p^3,$$

as required. (At $p = 0$ the point is the origin, a cusp; there the curve is $y = \pm 2(x/3)^{3/2}$, which is tangent to the x-axis $y = 0$, so the formula still holds.)

Where two tangents meet

For $p \neq q$, the tangents $y = px - p^3$ and $y = qx - q^3$ meet where

$$(p - q)\,x = p^3 - q^3 = (p - q)(p^2 + pq + q^2).$$

Since $p \neq q$ we may divide by $p - q$, giving $x = p^2 + pq + q^2$, and then

$$y = px - p^3 = p^3 + p^2q + pq^2 - p^3 = pq(p + q).$$

So the tangents meet at

$$(p^2 + pq + q^2,\ pq(p+q)).$$

The tangents have gradients p and q, so they are perpendicular exactly when $pq = -1$. Writing $u = p + q$, we have $p^2 + pq + q^2 = (p+q)^2 - pq = u^2 + 1$ and $pq(p+q) = -u$. Hence perpendicular tangents meet at $(u^2 + 1,\ -u)$, as required.

Ask yourself Does every point of $\widetilde{C}$ actually arise from a pair of perpendicular tangents?

Yes: given any real u, we need $p + q = u$ and $pq = -1$, so p and q are the roots of $z^2 - uz - 1 = 0$. Its discriminant $u^2 + 4$ is positive, so there are always two distinct real roots. So $\widetilde{C}$ is exactly the set of points from which two perpendicular tangents to C can be drawn.

Points on both curves

Eliminating the parameter, $\widetilde{C}$ is the parabola $x = y^2 + 1$. A point $(3t^2, 2t^3)$ of C lies on it exactly when

$$3t^2 = 4t^6 + 1 \quad \Longleftrightarrow \quad 4s^3 - 3s + 1 = 0, \quad \text{where } s = t^2.$$

Since $s = -1$ is a root, we can factorise:

$$4s^3 - 3s + 1 = (s+1)(4s^2 - 4s + 1) = (s+1)(2s-1)^2.$$

As $s = t^2 \geq 0$, the only possibility is $s = \frac{1}{2}$, so $t = \pm\frac{1}{\sqrt{2}}$. Then $x = 3t^2 = \frac{3}{2}$ and $y = 2t^3 = \pm\frac{1}{\sqrt{2}}$. The common points are

$$\left(\tfrac{3}{2},\ \tfrac{1}{\sqrt{2}}\right) \quad \text{and} \quad \left(\tfrac{3}{2},\ -\tfrac{1}{\sqrt{2}}\right).$$

Ask yourself What does the repeated factor $(2s-1)^2$ tell us?

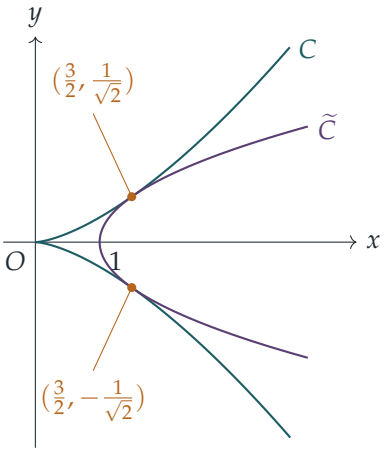
A double root usually signals tangency, and that is what happens here. At $t = \frac{1}{\sqrt{2}}$ the gradient of C is $t = \frac{1}{\sqrt{2}}$. On $\widetilde{C}$, $x = y^2 + 1$ gives $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{2y} = \frac{1}{2/\sqrt{2}} = \frac{1}{\sqrt{2}}$, the same value. By symmetry the same is true at the lower point. So the two curves touch at both points and do not cross.

The sketch

Eliminating t from C gives $y^2 = \frac{4}{27}x^3$, with $x \geq 0$. This is a semicubical parabola, symmetric in the x-axis, with a cusp at the origin pointing left. $\widetilde{C}$ is the parabola $y^2 = x - 1$, with vertex $(1, 0)$, opening to the right. To see which curve is outside the other, compare the two values of y^2 for $x \geq 1$:

$$f(x) = \tfrac{4}{27}x^3 - (x-1), \qquad f'(x) = \tfrac{4}{9}x^2 - 1,$$

so f has its minimum on $x \geq 1$ at $x = \frac{3}{2}$, where $f(\frac{3}{2}) = \frac{1}{2} - \frac{1}{2} = 0$. Hence $f(x) \geq 0$, with equality only at $x = \frac{3}{2}$. So $\widetilde{C}$ lies inside C (closer to the x-axis) and touches it at the two common points.



Checks

- Take $p = 1$, $q = -1$ (so $pq = -1$ and $u = 0$). The tangents are $y = x - 1$ and $y = -x + 1$, which are perpendicular and meet at $(1, 0)$, the vertex of $\widetilde{C}$, as $(u^2 + 1, -u)$ predicts.
- Take $p = 2$, $q = -\frac{1}{2}$, so $u = \frac{3}{2}$. The tangents $y = 2x - 8$ and $y = -\frac{1}{2}x + \frac{1}{8}$ meet where $\frac{5}{2}x = \frac{65}{8}$, so $x = \frac{13}{4} = u^2 + 1$ and $y = -\frac{3}{2} = -u$.
- The point $(1.5, 0.7071)$ satisfies both $y^2 = \frac{4}{27}x^3$ (since $\frac{4}{27} \cdot \frac{27}{8} = \frac{1}{2}$) and $y^2 = x - 1$. A computer check also confirmed that $3t^2 - 4t^6 - 1$ never goes above zero, so there are no other common points.

Common slips

- Forgetting that $s = t^2$ cannot be negative, and trying to use the root $s = -1$.
- Sketching the parabola crossing C at the common points instead of touching it there.

Beyond STEP

The curve $\widetilde{C}$ is the *orthoptic* of C: the locus of points from which two perpendicular tangents to C can be drawn; for a parabola it is the directrix. Describing a curve by its family of tangent lines leads to envelopes and to duality in projective geometry. Siklos, *Advanced Problems in Mathematics*, has more questions of this kind.

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